

ASCWG 2025 CTF Finals — Neon Maze Challenge Writeup

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Ascwg Finals

Ctf Writeup

Aspnet Security

Pentesting

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[\[image: Nour El Dien Bassiouny\]](#)

](https://medium.com/@NourBassiouny?source=post_page---byline--7cf426672a58-----)

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In the name of Allah, the Most Gracious, the Most Merciful { I only intend reform as much as I am able, and my success is only through Allah. In Him I put my trust, and to Him I turn.
}
```

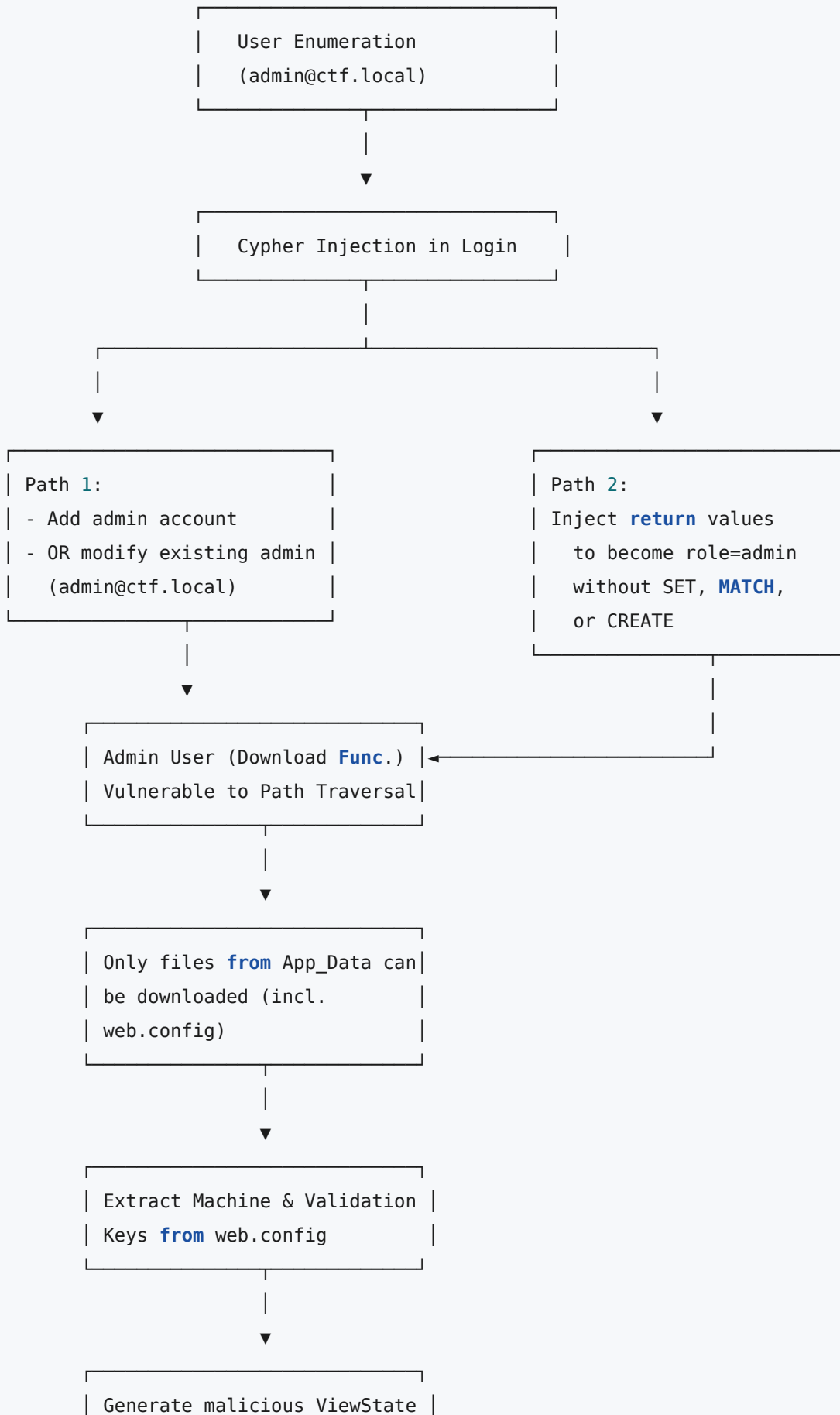
whoami?

I am **Nour El Dien Bassiouny**, Cyber Security Specialist at **iSec**, passionate about hacking and exploiting Applications & Infrastructures. The author of the **Neon Maze** web challenge in the **ASC War Games 2025**.

First of all, I would like to thank all the teams who participated in Arab Security Wargames. I hope you enjoyed it and learned something new.

My challenge was about **chaining multiple exploits together to achieve RCE**.

Here's a quick mindmap of how the challenge can be solved:



| to trigger RCE |

Before starting the writeup, let's go through some key terms to better understand the challenge.

What is Cypher?

- Cypher is **Neo4j's graph query language** that allows you to retrieve data from the graph. Think of it as SQL but for Graph databases.
- Originally designed for Neo4j, it was later opened up through the **openCypher** project and is now used in other databases like RedisGraph, Spark, Amazon Neptune, and SAP HANA Graph.

Graph vs Relational Databases

- **Data Model:** Relational databases use tables with rows and columns, while Graph databases use nodes and edges to represent entities and relationships.
- **Relationships:** In Relational databases, relationships are handled through joins. In Graph databases, relationships are stored directly, making traversals faster.
- **Use Cases:** Relational databases excel with structured, tabular data. Graph databases are better for highly connected data.

Basic Cypher Query

Here's a simple query to understand how Cypher works. Let's assume you have a `User` label, identified as `u`, with properties like username, email, age, first name, last name, and a hashed password.

```
CREATE (u:User {
  username: "Bassiouny1337",
  email: "test@gmail.com",
  fname: "Nour El Dien",
  lname: "Hisham Bassiouny",
  age: 20,
  password: "long_argon_hashed"
})
RETURN u.Email AS Email, u.Username AS Username
```

And a typical login query:

```
MATCH (u:User { Email:'test@gmail.com', Password:'hash_of_submitted_Password' })
RETURN u.Email AS Email, u.Username AS Username
```

There are many queries we could discuss, but this is enough for our challenge.

For more details, check this **Cypher Injection Cheatsheet**: <https://pentester.land/blog/cypher-injection-cheatsheet/>

How ViewState is Used

- The **ViewState** is generated by the server and sent back to the client as a hidden form field (`_VIEWSTATE`) during POST requests.
- The client sends it back to the server during post back. The server then deserializes it.
- ASP.NET uses different formatters (e.g., `ObjectStateFormatter` , `LOSFormatter` , `BinaryFormatter`) to serialize/deserialize objects.

Formatters

They convert data from one form to another. Example: `BinaryFormatter` serializes and deserializes objects or entire object graphs in binary format.

Gadgets

Classes that may allow code execution when processing untrusted data. Examples in .NET include `PSObject` , `TextFormattingRunProperties` , and `TypeConfuseDelegate` .

I think we've covered the **prerequisites for the NEON Maze challenge**. Now let's deep dive into it.

The application had 5 functions:

- Register
- Login
- Upload Photo
- Change Password
- Update Profile

By registering with a Gmail address, it tells us that it must be `@ctf.local` .

Register Function

By changing it to `ctf.local` , we can register successfully.

But wait, this might be a clue for something: user enumeration.

admin@ctf.local exists

By guessing popular usernames we found that `admin@ctf.local` exists.

we might think in something like Account takeover but how !?

After logging in it was observed that we have 2 juicy functions for account takeover: change password & Update Profile:

By testing and trying to takeover the admin account through them, all attempts failed. By checking the file upload function, it was found that it does all types of checks on the files,

even magic bytes.

So now, all we have is going back to the Login function. Maybe there is some type of injection.

Here is the normal login request with valid credentials:

But let's try to register with an email that has a single quote between its characters.

While trying to login, we found that we can't login.

So now login function seems suspicious to injection attacks.

By injecting 3 single quotes it throws errors and here what make it a bit hard to be found that to throw errors 2 single quotes must be injected at the beginning of the exploit payload.

From the error it is obvious that login is vulnerable to cypher injection and that leaks the whole cypher query too

How to exploit ?

There is 2 paths to exploit

- Add an admin account or change the password of the enumerated admin account.
- Modify return values (this is the easy way to exploit, because `SET`, `MATCH`, and `CREATE` are filtered).

So let's take the first path and see how to bypass this filtering by using the APOC plugin in our payload.

1. Change admin's password via cypher injection

```
{
  "Email": "a'}) WITH reduce(m='',c IN ['M','A','T','C','H'] | m+c) AS m,
    reduce(s='',c IN ['S','E','T'] | s+c) AS s,
    reduce(r='',c IN ['R','E','T','U','R','N'] | r+c) AS r
  CALL apoc.cypher.runWrite(
    m + ' (n:User) WHERE n.Email=\\'admin@ctf.local\\' ' +
    s + ' n.Password=\\'pwned123\\' ' +
    r + ' n.Username AS Username, n.Role AS Role',
    {}
  ) YIELD value
  RETURN value.Username AS Username, value.Role AS Role //",
  "Password": "irrelevant"
}
```

For Validation let's try to login with the new credentials

2. Create admin account via cypher injection

For Validation let's try to login with the new credentials.

3. Escalate Normal user (bassiouny@ctf.local) to admin user

```
{
  "Email": "bassiouny@ctf.local'}) RETURN u.Username AS Username,'admin' AS Role //",
  "Password": "irrelevant"
}
```

and by pasting response in our browser and checking profile page, Boom our user become admin .

By checking admin page we found download logs function:

As seen below, the Download Logs button and the request show how it downloads the file.

By checking the response headers, we found that the web application is ASP.NET, and by searching we identified a general project file structure.

```
YourWebApp/
|
├─ App_Data/                # Used to store local DB files (.mdf), XMLs, etc. (NOT
served by IIS)
├─ App_Start/                # MVC only: RouteConfig.cs, FilterConfig.cs,
BundleConfig.cs, etc.
├─ App_GlobalResources/     # For .resx files (global resources like localization)
├─ App_LocalResources/      # Page-specific resources (.resx)
├─ bin/                     # Compiled assemblies (DLLs)
├─ Content/                 # CSS, images, fonts
├─ Controllers/             # MVC: C# controller classes (e.g., HomeController.cs)
├─ Models/                  # C# classes representing data models
├─ Views/                   # MVC: Razor views (*.cshtml)
|   └─ Shared/              # Shared layout views (_Layout.cshtml,
_ValidationScriptsPartial.cshtml)
├─ Scripts/                 # JavaScript files, jQuery, etc.
├─ Web.config               # Main configuration file (VERY sensitive)
├─ Global.asax              # Application start, error handling
├─ Default.aspx            # Web Forms: Entry point (if Web Forms app)
├─ packages.config          # NuGet package listing (older apps)
└─ .csproj / .sln          # Visual Studio project/solution files (not deployed, but
important)
```

So now our target is to read the `web.config` file, which is very sensitive and contains machine keys and possibly credentials for other services.

or

After leakage of `validationKey` & `decryptionKey`, we needed to find a place to send a malicious object. After some searching, it was found that the ViewState is sent in the Update Profile request body.

So now we need to craft our exploit using ysoserial.net.

ASP.NET makes use of `LosFormatter` to serialize the viewstate and send it to the client as the hidden form field. Once the serialized viewstate is sent back to the server during a POST request, it gets deserialized using `ObjectStateFormatter`.

Then send the request with the crafted ViewState.

The Flag Was: **ASCWG{N30 Gr4ph!nj3c7 LF! V!3w\$T@84}**

I hope you guys enjoyed the challenge and the writeup.

Feel free to contact me on [LinkedIn](#) or [twitter](#).

And peace be upon you, and the mercy of Allah and His blessings.

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